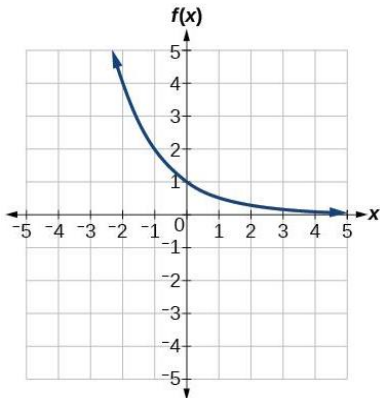
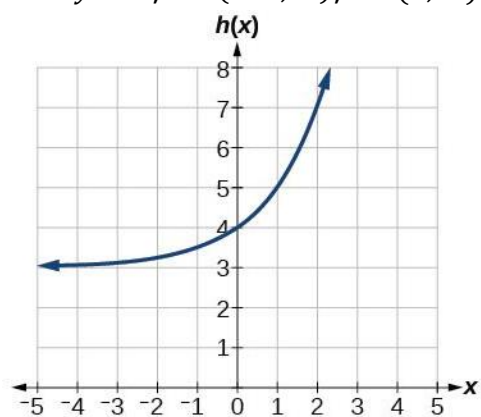


Section 4.1 – 4.2 Exercise Answers

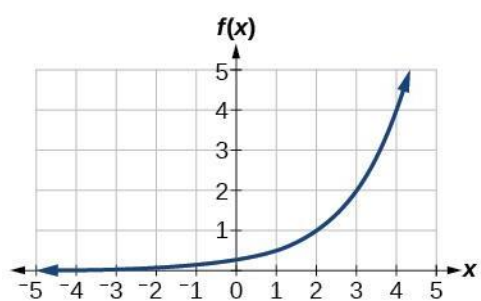
1. a^4
2. m^3n^2
3. b^6c^8
4. $x^{15}y^{10}$
5. $\frac{1}{x^5}$
6. m^4
7. 1
8. $\frac{q^5}{p^6}$
9. $\frac{y}{81z^6}$
10. Neither
11. Exponential Growth
12. Neither
13. Exponential Decay
14. \$104,053.65
15. \$669.42
16. 0.016
17. -4
18. 20.0855
19. -0.2707
20. 174.3
21. 483.8146
22. 1.3333
23. \$35,838.76
24. \$82,247.78; \$449.75
25. \$7,353.84; \$7,469.63; \$7,496.71; \$7,510.44
26. HA: $y = 0$; D: $(-\infty, \infty)$; R: $(0, \infty)$



27. HA: $y = 3$; D: $(-\infty, \infty)$; R: $(3, \infty)$



28. HA: $y = 0$; D: $(-\infty, \infty)$; R: $(0, \infty)$



29. $f(x) = 4^x + 4$

30. $f(x) = 4^x - 3$

31. $f(x) = 4^{x+2}$

32. $f(x) = 4^{x-5}$

33. $f(x) = -4^x$

34. $f(x) = 4^{-x}$